

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8395**

AY: 2025-26	Session: Odd MSE October 2025	Date: 08-10-2025
Sem: 1	Slot: 10:00am - 11:00am	C. Code: CS413
Exam Type: Written	Block: 305	Seat No: 2023701002
Course Section: 1		

L_41_683_986_L_1505_41356



Name of the candidate. Shrikant Gopinath Kerbade

Candidate's UID number:

2	0	2	3	7	0	1	0	0	2
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Examination class room number:

3	0	5
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Invigilator's signature with date:

 8/10

(To be filled in by the candidate)

EXAMINATION: BE
(For example FY MCA/FY BTECH)

SEMESTER : 7th

PROGRAM: CSE (DS)

I / II / III / IV / V / VI / VII / VIII
7

DAY AND DATE: 8/10/2025

COURSE NAME: Deep Learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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1 q] →

$$\text{let } f(w) = \frac{1}{2} \|Xw - y\|^2$$

gradient update rule :-

$$\theta_{\text{new}} = \theta_{\text{old}} - \eta \nabla L(\theta_{\text{old}})$$

where

θ can be w, b

η is learning rate

∇L is a loss or gradient

gradient (Partial derivatives)

$$\frac{\partial}{\partial w} f(w) = \frac{1}{2} (Xw - y) \cdot (w - y)$$

The learning rate η should be greater than 0 and less than $\frac{2}{\lambda_{\text{max}}}$.

$X^T X$ is a covariance matrix.



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16] 1

$$SSF = \sum_{i=1}^N (\hat{y}_i - \bar{y}_i)^2$$

The Squared errors of values will be very diverse cause of making square of error.

So large value differences creates high variance which leads to overfit model.

example: -

Having good variation like in Mean Squared Error method scales down the value which makes ~~low~~ low variance or balanced variance.

This framework will not distribute values normally or it gives very high differences which affects generalization.

high variance - overfitting
high bias - underfitting

so we have to maintain a normalised balance of both to have a good or accurate model.

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$$H \times W \times C \text{ with } K, F \times F$$

~~XXXXXXXXXX~~

Input Size = $H \times W \times C$ 2D
no of filters = K
filter size = $F \times F$

so $(H \times W \times C) \times KCF$

if padding P added then $2P$.

[illegible]

$Qp + H \times W \times C$ with $K \in$

not padding then

[HWCKE] will be complexity.

for parameter efficiency

$$(K_w \times K_h \times C_{in}) \times C_{out}$$

By using this we can check for parameter efficiency.

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25] →

Linear undercomplete autoencoder

Encoder

$$h = Wx \quad \dots (1)$$

Decoder

$$\hat{x} = Dh$$

by (1)

$$\hat{x} = DWx$$

$$D = W^T$$

then

$$\hat{x} = W^T W x \quad \dots (2)$$

$$MSE = \frac{1}{N} \sum_{i=1}^N \|x - \hat{x}\|^2$$

where

PCA

$$z = Wx \quad \dots (3)$$

W is eigenvector of $X^T X$ covariance matrix.

by (2) reconstruct the $\hat{x}$.

$$\hat{x} = W^T z$$

by (3) Put $z = Wx$

$$\hat{x} = W^T W x \quad \dots (4)$$

$$W^T W x \approx W^T W x$$

So yes linear under complete autoencoder



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		tries PCA to reconstruction.
		Overcomplete regularization
		Overcomplete
		Overcomplete Autoencoder without regularization
		it tries to copy as it is and makes it identical
		Solution is use regularized Autoencoders.
		encoder $h = wx$
		$h = wx = x$ Nothing improved. identical.

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3a) 1

$$\frac{\partial L}{\partial h_{t-k}}$$

LSTM has cell state update which maintains the memory and solves the vanishing problem.

Cell state maintains a long-term memory such that the finding gradient expression $\frac{\partial L}{\partial h_{t-k}}$

for long doesn't occurs like in RNN.

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$$

This is the cell state update equation from LSTM which maintains the long term memory efficiently.

$$f_t = \sigma(W_f[h_{t-1}, x] + b_f)$$

This equation is of forget gate in LSTM which removes not important data from long-term memory and modifies

$\frac{\partial L}{\partial h_{t-k}}$ recursion in RNNs.



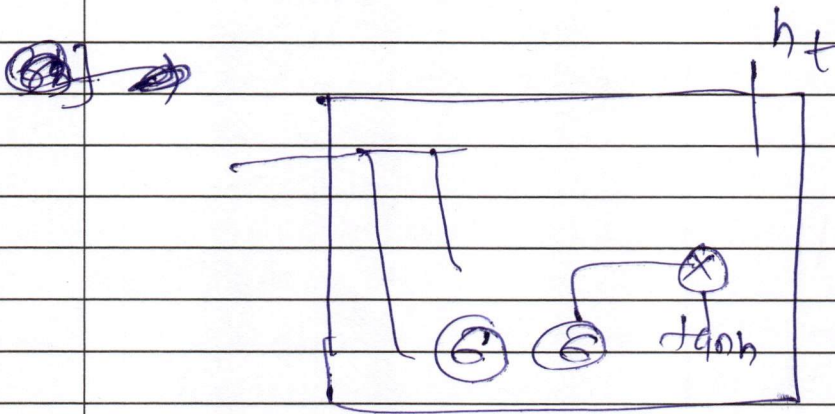
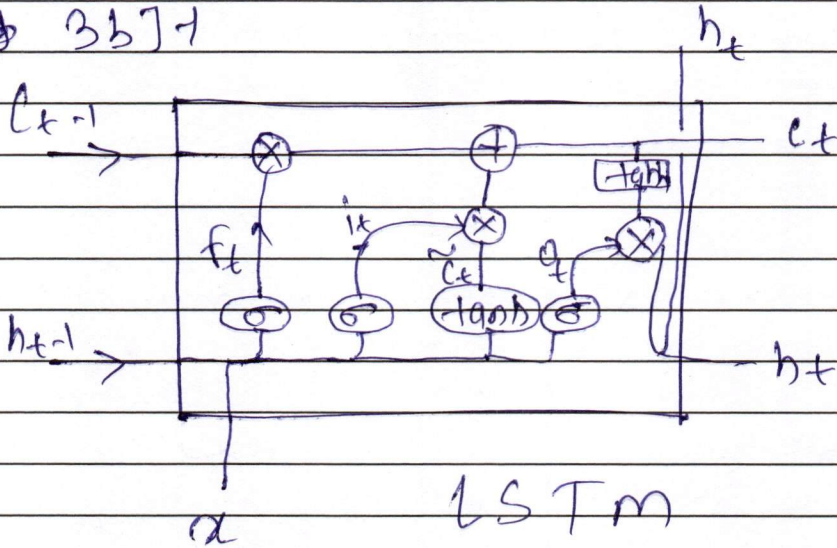
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Sr. No. 8395



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Sr. No.

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$$\frac{N - K + 2P}{S} + 1$$

$$(K_h \times K_w \times C_{in}) \times C_{out}$$